

Life Mathematics (8 th std)

Choose the Appropriate Answer

1.If 30 % of x is 150, then x is _____.

- (a) 600 (b) 300 (c) 450 (d) 500

Ans: (d) 500

$$x = \frac{150 \times 100}{30} = 500$$

2.2 minutes is _____ % to an hour.

- (a) $3 \frac{2}{3}$ % (b) $3 \frac{1}{3}$ % (c) $3 \frac{1}{2}$ % (d) None

Ans: (b) $3 \frac{1}{3}$ %

$$1 \text{ hr} = 60 \text{ min} \Rightarrow x\% = \frac{2}{60} \times 100 = \frac{200}{60} = \frac{10}{3} = 3 \frac{1}{3} \%$$

3.If x% of x = 25, then x = _____.

- (a) 60 (b) 55 (c) 50 (d) 65

$$\frac{x}{100} \times x = 25 \Rightarrow x^2 = 25 \times 100 = 2500 \Rightarrow x = \sqrt{2500} = 50$$

4.In a school of 1400 students, there are 420 girls. The percentage of boys in the school is _____.

- (a) 80 % (b) 75 % (c) 70% (d) 65%

Ans: (c) 70 %

$$\begin{aligned} \text{Number of boys in school} &= 1400 - 420 = 980 \\ \text{\% of boys in school} &= \frac{\text{Number of boys}}{\text{Total number of students}} \times 100 = \frac{980}{1400} \times 100 = 70\% \end{aligned}$$

5.0.5252 is _____ %.

- (a) 5.252 (b) 52.52 (c) 0.5252 (d) 525.2

Ans: (b) 52.52%

$$0.5252 \times 100 = 52.52\%$$

6.12% of 250 litre is the same as _____ of 150 litre.

- (a) 10% (b) 15% (c) 20% (d) 30%

Ans: (c) 20%

$$\begin{aligned} 12\% \text{ of } 250 &= \frac{12}{100} \times 250 = 30 \text{ lit} \\ \text{Percentage: } &\frac{30}{150} \times 100 = 20\% \end{aligned}$$

7. If three candidates A, B and C in a school election got 153, 245 and 102 votes respectively, then the percentage of votes got by the winner is _____.
 (a) 48% (b) 49% (c) 50% (d) 45%

Ans: (b) 49%

$$\begin{aligned} \text{Total votes} &= 1 + 2 + 3 = 153 + 245 + 102 = 500 \\ \text{\% of votes for winner} &= \frac{245}{500} \times 100 = 49\% \end{aligned}$$

8. 15% of 25% of 10000 = _____.
 (a) 375 (b) 400 (c) 425 (d) 475

Ans: (a) 375

$$\frac{25}{100} \times 10000 = 2500 \Rightarrow \frac{15}{100} \times 2500 = 375$$

9. When 60 is subtracted from 60% of a number to give 60, the number is
 (A) 60 (b) 100 (c) 150 (d) 200

Ans: (d) 200

$$\frac{60}{100} \times x - 60 = 60 \Rightarrow x = 120 \times \frac{100}{60} = 200$$

10. If 48% of 48 = 64% of x, then x =
 (a) 64 (b) 56 (c) 42 (d) 36

Ans: (d) 36

$$\frac{48}{100} \times 48 = \frac{64}{100} \times x \Rightarrow x = \frac{48^3 \times 48^{12}}{64_{21}} = 36$$

11. Loss or gain percentage is always calculated on the _____.
 (a) Selling Price (b) Cost Price (c) Discount (d) None

Ans: (b) Cost Price

12. A fruit vendor sells fruits for ₹ 200 gaining ₹ 40. His gain percentage is
 (a) 20% (b) 22% (c) 25% (d) 16

Ans: (c) 25%

$$\begin{aligned} \text{CP} - \text{Selling price} - \text{gain} &= 200 - 40 = 160 \\ \text{Gain \%} &= \frac{\text{Gain}}{\text{CP}} \times 100 = \frac{40}{160} \times 100 = 25\% \end{aligned}$$

13. By selling a flower pot for ₹ 528, a woman gains 20%. At what price should she sell it to gain 25%?

(a) ₹ 500 (b) ₹ 550 (c) ₹ 553 (d) ₹ 573

Ans: (b) ₹ 550

14. A man buys an article for ₹ 150 and makes overhead expenses which are 12% of the cost price. At what price must he sell it to gain 5%?

- (a) ₹ 180 (b) ₹ 168 (c) ₹ 176.40 (d) ₹ 88.20

Ans: (c) ₹ 176.40

15. What is the marked price of a hat which is bought for ₹ 210 at 16% discount?

- (a) ₹ 243 (b) ₹ 176 (c) ₹ 230 (d) ₹ 250

Ans: (d) ₹ 250

16. The single discount in % which is equivalent to two successive discounts of 20% and 25% is

- (a) 40% (b) 45% (c) 5% (d) 22.5%

Ans: (a) 40%

17. The compound interest on ₹ 5000 at 12% p.a for 2 years, compounded annually is _____.

- (a) ₹ 1270 (b) ₹ 1275 (c) ₹ 1272 (d) ₹ 1278

Ans: (b) ₹ 1272

18. The number of conversion periods in a year, if the interest on a principal is compounded every two months is _____.

- (a) 2 (b) 4 (c) 6 (d) 12

Ans: (c) 6

19. The time taken for ₹ 4400 to become ₹ 4851 at 10%, compounded half yearly is _____.

- (a) 6 months (b) 1 year (c) 112 years (d) 2 years

Ans: (b) 1 year

20. The cost of a machine is ₹ 18000 and it depreciates at 16.23% annually. Its value after 2 years will be _____.

- (a) ₹ 2000 (b) ₹ 12500 (c) ₹ 15000 (d) ₹ 16500

Ans: (b) ₹ 12500

21. The sum which amounts to ₹ 2662 at 10% p.a in 3 years, compounded yearly is _____.

- (a) ₹ 2000 (b) ₹ 1800 (c) ₹ 1500 (d) ₹ 2500

Ans: (a) ₹ 2000

22. The difference between compound and simple interest on a certain sum of money for 2 years at 2% p.a is U. The sum of money is _____.

- (a) ₹ 2000 (b) ₹ 1500 (c) ₹ 3000 (d) ₹ 2500

Ans: (d) ₹ 2500

23. A can finish a job in 3 days whereas B finishes it in 6 days. The time taken to complete the job working together is _____ days.

- (a) 3 days (b) 4 days (c) 2 days (d) 1 day

Ans: (c) 2 days

24.If 5 persons can do 5 jobs in 5 days, then 50 persons can do 50 jobs in _____ days.

- (a) 3 (b) 4 (c) 5 (d) 6

Ans: (c) 5

25.**A** can do a work in 24 days. If **A** and **B** together can finish the work in 6 days, then B alone can finish the work in _____ days.

- (a) 6 (b) 8 (c) 7 (d) 9

Ans: (b) 8

26.**A** alone can do a piece of work in 35 days. If **B** is 40% more efficient than **A**, then **B** will finish the work in _____ days.

- (a) 25 (b) 27 (c) 26 (d) 28

Ans: (a) 25

27.**A** alone can do a work in 10 days and **B** alone in 15 days. They undertook the work for ₹ 200000. The amount that **A** will get is

- (a) 1,50,000 (b) 1,30,000 (c) 1,20,000 (d) 1,25,000

Ans: (c) ₹ 1, 20,000

28.What percentage one day is 10 hours?

- (a) 40 (b) 41 (c) 42 (d) 43

Ans: (b) 41

$$\text{Percentage} = \frac{10}{24} \times 100 = 41$$

29.If the selling price of an article is less than the cost price of the article, then there is a _____.

- (a) Gain (b) Loss (c) Discount (d) None

Ans: (b) Loss

30.The formula to find the simple interest for a given principal is _____.

- (a) $PN / R \times 100$ (b) $100 / PNR$ (c) $PNR / 100$ (d) None

Ans: (c) $PNR / 100$

31.On what a discount is calculated?

- (a) S.P (b) C.P (c) Marked Price (d) None of these

Ans: (c) Marked Price

32.On which figure the VAT of a product is calculated?

- (a) S.P (b) C.P (c) Marked Price (d) None of these

Ans: (a) S.P

33.On which of the following percent profit or profit loss is calculated?

- (a) S.P (b) C.P (c) Marked Price (d) None of these

Ans: (b) C.P

34.If an article sold for Rs.100 then there is a gain of Rs.20, which of the following is the gain percent?

- (a) 25% (b) 22% (c) 20% (d) 16%

Ans: (a) 25%

35.An article is at 10% more than the CP. If discount of 10% is allowed then which of the following is right?

- (a) 1% gain (b) 1% loss (c) No gain no loss (d) 1.1% loss

Ans: (b) 1% loss

36.A building worth Rs.1 is depreciated by R% per annum. Which of the following is true?

- (a) $P [1 - 5/100]$ (b) $P [1 + 5/100]$
(c) $P [(1 + 5/100) - 1]$ (d) $P [1 - (1 - 5/100)]$

Ans: (a) $P [1 - 5/100]$

37.If M.P of a box is Rs.10 and a discount of 10% is allowed then what should be the sale price?

- (a) Rs.10 (b) Rs.9 (c) Rs.11 (d) None of these

Ans: (b) Rs.9

38.What should be the rate of interest per annum if interest is calculated quarterly?

- (a) Reduced to half (b) Reduced to one fourth
(c) Is doubled (d) Becomes four times

Ans: (b) Reduced to one fourth

39.What time period is taken when interest is calculated half yearly?

- (a) Twice as much as the number of given years
(b) Half as much as the number of given years
(c) Same as the number of given years
(d) None of these

Ans: (a) Twice as much as the number of given years

40.What should be percentage gain on a product when it is sold for Rs.120 with a gain of Rs.20?

- (a) 20% (b) 25% (c) 22% (d) 16.25%

Ans: (a) 20%

41.If 'x' and 'y' are in a direct proportion then which of the following is correct?

- (a) $x - y = \text{constant}$ (b) $x + y = \text{constant}$
(c) $x \times y = \text{constant}$ (d) $x / y = \text{Constant}$

Ans: (d) $x / y = \text{Constant}$

42.If 'x' and 'y' are in an inverse variation then which of the following is correct?

- (a) $x - y = \text{constant}$ (b) $x + y = \text{constant}$
(c) $x y = \text{constant}$ (d) $x / y = \text{Constant}$

Ans: (b) $x + y = \text{constant}$

43. If 'A' can finish a work in 'n' days then part of work finished in 1 day is:
(a) $1 - n$ (b) $1 / n$ (c) $n - 1$ (d) None of these

Ans: (b) $1 / n$

44. If amount of work completed by 'A' in one day is $1 / n$ then the whole work will be finished by 'A' is:

(a) n days (b) $1 - n$ days (c) $n - 1$ days (d) None of these.

Ans: (a) n days

45. If an increase in one quantity brings about a corresponding decrease in the other and vice versa, then the two quantities vary:

(a) Directly (b) Sometimes directly and sometimes inversely

(c) Inversely (d) None of these

Ans: (c) Inversely

46. "If speed is more then time to cover a fixed distance would be less". This is a case of:

(a) Inverse variation (b) Direct variation

(c) Joint variation (d) None of the above

Ans: (a) Inverse variation

47. Which of the following is not a case of direct variation?

(a) Number of sheets of some kind are increased when their total weight is increased

(b) More quantity of petrol is required to travel more distance with a fixed speed

(c) More fees would be collected if number of students increased in a class

(d) Time taken will be less if number of workers are increased to complete the same work.

Ans: (d) Time taken will be less if number of workers are increased to complete the same work.

48. Which of the following is a case of direct variation;

(a) If the length of radius is increased the circumference will be increased

(b) If number of students in a hostel are increased then the fixed food provision will last for less days

(c) For fixed duration, more the periods, lesser will be the duration of one period

(d) In case of a cylindrical vessel, lesser the diameter more is the level of water in it.

Ans: (a) If the length of radius is increased the circumference will be increased

49. If x and y vary inversely. Then using the following table?

x	5
y	30

The value of x for y = 10 is?

(a) 10 (b) 40 (c) 15 (d) 20

Ans: (c) 15

50. The ratio of speed of cycle 12 km per hour to the speed of scooter 36 km per hour is

- (a) 1:2 (b) 1:3 (c) 1:4 (d) None of the above

Ans: (b) 1:3

51. The ratio of 10m to 10 km is:

- (a) 1/10 (b) 1/100 (c) 1/1000 (d) 1000

Ans: (c) 1 / 1000

52. The percentage of 1:4 is:

- (a) 75% (b) 50% (c) 25% (d) 100%

Ans: (c) 25%

53. The percentage of 2:5

- (a) 20% (b) 50% (c) 60% (d) 40%

Ans: (d) 40%

54. If 50% of students are good at science out of 20 students. Then the number of students good at science is:

- (a) 10 (b) 15 (c) 5 (d) 11

Ans: (a) 10

55. The price of a motorcycle was Rs.34, 000 last year. It has increased by 20% this year. The price of motorcycle now is:

- (a) Rs.36, 000 (b) Rs.38, 800 (c) Rs.40, 800 (d) Rs.32, 000

Ans: (c) Rs.40, 800

$20\% \text{ of Rs.34, 000} = 20/100 \times 34,000 = \text{Rs.6800}$ $\text{New price} = \text{Rs.34, 000} + \text{Rs.6800} = \text{Rs.40, 800}$
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56. An item marked at Rs.840 is sold for Rs.714. The discount % is:

- (a) 10% (b) 15% (c) 20% (d) 25%

Ans: (b) 15%

$\text{Discount} = \text{Marked Price} - \text{Sale Price} = 840 - 714 = \text{Rs.126}$ $\text{Discount \%} = (126/840) \times 100\% = 15\%$
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57. A person got a 10% increase in his salary. If his salary was Rs.50000, then the new salary is:

- (a) Rs.55000 (b) Rs.60000 (c) Rs.45000 (d) Rs.65000

Ans: (a) Rs.55000

58. The cost of the article was Rs.15500 and Rs.500 was spent on its repairing. If it is sold for a profit of 15%. The selling price of the article is:

- (a) Rs.16400 (b) Rs.17400 (c) Rs.18400 (d) Rs.19400

Ans: (c) Rs.18400

$\text{Total cost} = 15500 + 500 = 16000$ $\text{Profit} = (\text{Profit\%} \times \text{Cost price})/100 = (15 \times 16000)/100 = 2400$ $\text{Selling Price} = \text{Profit} + \text{cost price} = 2400 + 16000 = \text{Rs.18400}$

59. Wahid bought an air cooler for Rs.3300 including a tax of 10%. The price of the air cooler before VAT was added is:

- (a) Rs.2000 (b) Rs.3000 (c) Rs.2500 (d) Rs.2800

Ans: (b) Rs.3000

$$\text{original price} = \text{Rs. } (100/110) \times 3300 = \text{Rs. } 3000$$

60. The ratio of 50 paise to Rs.1 is:

- (a) 1: 2 (b) 2: 1 (c) 1: 1 (d) 1: 5

Ans: (a) 1: 2

61. The Ratio of 10m to 1 Km is Equal To:

- (a) 1/10 (b) 1/100 (c) 1/1000 (d) 1000

Ans: (b) 1 / 100

62. The ratio of 1m and 100 cm is:

- (a) 1:100 (b) 1:10 (c) 10:1 (d) 1:1

Ans: (d) 1:1

63. The percentage of ratio 2:3 is:

- (a) 44.44% (b) 55.55% (c) 66.66% (d) None of the above

Ans: (c) 66.66%

$$\frac{2}{3} \times 100\% = 200/3 = 66.66\ldots\%$$

64. The ratio 4: 25 converted to percentage is:

- (a) 8% (b) 4% (c) 16% (d) 25%

Ans: (c) 16%

65. Convert the fraction $\frac{1}{8}$ into a percentage.

- (a) 12.5 % (b) 25% (c) 8% (d) 16%

Ans: (a) 12.5%

66. The ratio of speed of a motorbike 50 km per hour and speed of cycle 20 km per hour is:

- (a) 2:5 (b) 5:1 (c) 5:2 (d) 1:2

Ans: (c) 5:2

67. If 40% of students passed in the Mathematics test, out of 40 students in the class. What is the number of students who have passed?

- (a) 10 (b) 16 (c) 5 (d) 11

Ans: (b) 16

68. 25% of 60 students of a class likes to play football. How many students does not like to play football?

- (a) 45 (b) 15 (c) 25 (d) 50

Ans: (a) 45

69. Out of 40 students, 25% passed in a class. How many students did actually pass?

- (a) 10 (b) 20 (c) 30 (d) 40

Ans: (a) 10

70. If $x = 20$ and $y = 40$, then x and y are:

- (a) Directly proportional (b) Inversely proportional
(c) Neither directly nor inversely proportional (d) Cannot be determined

Ans: (a) Directly proportional

71. If $x = 15$ and $y = 1/30$, then x and y are:

- (a) Directly proportional (b) Inversely proportional
(c) Neither directly nor inversely proportional (d) Cannot be determined

Ans: (b) Inversely proportional

72. If $x \propto y$ and $x_1 = 5$, $y_1 = 210$ and $x_2 = 2$, then find y_2 ?

- (a) 200 (b) 84 (c) 99 (d) 70

Ans: (b) 84

73. If the weight of 12 sheets of thick paper is 40 grams, how many sheets of the same paper would weigh 2500 grams?

- (a) 750 (b) 800 (c) 850 (d) 950

Ans: (a) 750

$$12/40 = x/2500 \Rightarrow x = (12 \times 2500)/40 \Rightarrow x = 750$$

74. The scale of a map is given as 1:300. Two cities are 4 km apart on the map. The actual distance between them is:

- (a) 1000 km (b) 1100 km (c) 1200 km (d) 1300 km

Ans: (c) 1200 km

75. If x and y are inversely proportional, then which one is true?

- (a) $x_1/y_1 = x_2/y_2$ (b) $x_1/x_2 = y_1/y_2$ (c) $x_1/x_2 = y_2/y_1$ (d) $x_1 \cdot x_2 = y_1 \cdot y_2$

Ans: (c) $x_1/x_2 = y_2/y_1$

76. 6 pipes are required to fill a tank in 1 hour 20 minutes. If we use 5 such types of pipes, how much time it will take to fill the tank?

- (a) 120 minutes (b) 96 minutes (c) 80 minutes (d) 85 minutes

Ans: (b) 96 minutes

77. If 12 workers can build a wall in 50 hours, how many workers will be required to do the same work in 40 hours?

- (a) 10 (b) 13 (c) 14 (d) 15

Ans: (d) 15

78. A man walks 20 km in 5 hours. How much time it will take for him to walk 32 km?

- (a) 3 hours (b) 4 hours (c) 6 hours (d) 8 hours

Ans: (d) 8 hours

79.If 300 Kg of coal cost 6000₹, then find the cost of 120 kg of coal?

- (a) 1200₹ (b) 2400₹ (c) 3200₹ (d) 4200₹

Ans: (b) 2400₹

80.Manic types 200 words in 30 minutes. How many words she will type in 12 minutes?

- (a) 100 (b) 80 (c) 60 (d) 50

Ans: (b) 80

81.If it takes 40 days for 120 men to complete a work, how long will it take for 80 men to complete the same work?

- (a) 50 days (b) 60 days (c) 80 days (d) 100 days

Ans: (b) 60 days

82.6 pipes are required to fill a tank in 80 minutes. How long will it take if only 5 pipes of the same type are used?

- (a) 96 minutes (b) 102 minutes (c) 106 minutes (d) 108 minutes

Ans: (a) 96 minutes

83.If 12 labours can construct a road in 50 hours, how many labours will be required to construct the same road in 40 hours?

- (a) 10 (b) 12 (c) 14 (d) 15

Ans: (d) 15

84.If 100 students took 20 days to clean a garden. How many days it will take to clean the garden if 25 more students are added?

- (a) 10 (b) 12 (c) 16 (d) 18

Ans: (c) 16

85.A car takes 18 hours to ride 720 kilometres. Time taken by the car to travel 360 kilometres is:

- (a) 10 hours (b) 11 hours (c) 9 hours (d) 16 hours

Ans: (c) 9 hours

86.The perimeter of a square and its side is in:

- (a) Direct Proportion (b) Indirect Proportion
(c) Neither direct nor indirect (d) Cannot be determined

Ans: (a) Direct Proportion

87.If the cost price of a book is ₹120 and it is sold for ₹150, what is the profit percentage?

- (a) 10% (b) 20% (c) 5% (d) 30%

Ans: (c) 25%

88.What is the simple interest on ₹2000 at a rate of 5% per annum for 3 years?

- (a) ₹200 (b) ₹300 (c) ₹400 (d) ₹500

Ans: (b) ₹300

89.If $x \propto 1/y$ and $y = 2$, then when $x = 5$, y is:

- (a) 1 (b) 2 (c) 5 (d) 10

Ans: (a) 1

90.If x and y vary directly, and $x = 6$ when $y = 2$, find y when $x = 3$:

- (a) 1 (b) 2 (c) 4 (d) 6

Ans: (a) 1

91.Anita types 540 words in half an hour. How many words she can type in 120 seconds.

- (a) 3 (b) 12 (c) 36 (d) 28

Ans: (c) 36

$$540 / 1800 = a / 120 \Rightarrow 540 \times 120 = 1800 \times a \Rightarrow a = 36$$

92.Cost of 5m carpet is Rs.270/- . Then what will be the cost of a 12m carpets?

- (a) 624 (b) 684 (c) 628 (d) 648

Ans: (d) 648

Meter	5m	12m
RS.	270	?

Direct proportion = $(270 \times 12) / 5 = 648$

93.A train covers the distance in 2 hrs at the speed of 80 km/hr. Then how much time will it take to cover the same distance at speed the speed of 90 km/hr.

- (a) 1 hour 46 minutes (b) 1 hour 30 minutes
(c) 1 hour 50 minutes (d) 1 hour 40 minutes

Ans: (a) 1 hour 46 minutes

Speed	80 km/ hr	90 km / hr
Time	120 mins	?

Inverse Proportion = $x_1 y_1 = k x_2 \Rightarrow k = (80 \times 120) / 90 = 106.66 \text{ hrs} \Rightarrow 1 \text{ hr } 46 \text{ mins}$

94.Mani types 200 words in 30 minutes. Find how many words she will type in 12 minutes?

- (a) 100 (b) 80 (c) 60 (d) 50

Ans: (b) 80

$$30/200 = 12/x \Rightarrow x = 12 \times 200/30 \Rightarrow x = 80$$

95.It takes 40 days for 120 men to complete a work. How long will it take for 80 men to complete the same work?

- (a) 50 days (b) 60 days (c) 80 days (d) 100 days

Ans: (b) 60 days

$$40 \times 120 = 80 \times x \Rightarrow x = 60$$

96.If 6 pipes are required to fill a tank in 80 minutes. Then find how long will it take if only 5 pipes of the same type are used?

- (a) 96 minutes (b) 102 minutes (c) 106 minutes (d) 108 minutes

Ans: (a) 96 minutes

$$80 \times 6 = x \times 5 \Rightarrow x = 96$$

97.A man walks 20 km in 5 hours. Find out the time it will take for him to walk 32 km?

- (a) 3 hours (b) 4 hours (c) 6 hours (d) 8 hours

Ans: (d) 8 hours

$$20/5 = 32/x \Rightarrow x = 32/4 = 8$$

98.The Price of a Motorcycle Was Rs.34, 000 Last Year. It Has Increased by 20% This Year. the Price of Motorcycle Now Is:

- (a) Rs.36,000 (b) Rs.38,800 (c) Rs.40,800 (d) Rs.32,000

Ans: (c) Rs.40, 800

$$20 / 100 \times 34,000 = \text{Rs.}6800$$
$$\text{New price} = \text{Rs.}34, 000 + \text{Rs.}6800 = \text{Rs.}40, 800$$

99.An Item Marked at Rs.840 Is Sold for Rs.714. The Discount % is:

- (a) 10% (b) 15% (c) 20% (d) 25%

Ans: (b) 15%

$$\text{Discount} = \text{Marked Price} - \text{Sale Price} = 840 - 714 = \text{Rs.}126$$
$$\text{Discount \%} = (126 / 840) \times 100\% = 15\%$$

100.A Person Got an Increase of 10% in His Salary. If His Salary Was Equal to Rs.50000, Then the New Salary Is:

- (a) Rs.55000 (b) Rs.60000 (c) Rs.45000 (d) Rs.65000

Ans: (a) Rs.55000

$$10\% \text{ of Rs.}50000 = (10 / 100) \times 50000 = \text{Rs.}5000$$
$$\text{New salary} = \text{Rs.}50000 + \text{Rs.}5000 = \text{Rs.}55000/-$$